

Load Interconnection System Impact Study Report

For Queue Project Q20-1454

Mickens 345 kV
January 20, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1454. The Interconnection Customer proposes a 204.3 MW large load (data center) facility interconnecting at the Mickens 345 kV substation in the Virginia area. The request was received on April 12, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 414 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 24 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,820,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
TOANO 138/1kV Autotransformer	MIRA VISTA-YARNEN 345 KV	119.1	579	34.2
DANC LFGTE FACILITY-ELKO 138 KV	YARNEN-MIRA VISTA 345 KV	118.4	282	9.9
MCCARTNEY-YARNEN 345 KV line	FYFFE 345 kV/138 kV Autotransformer	116.3	1,656	50.3
FYFFE 345/138 kV TX	MCCARTNEY-YARNEN 345 KV	113.1	949	28.0
DANC LFGTE FACILITY-BONNY BROOK 138 KV	MIRA VISTA-YARNEN 345 KV	112.6	282	9.9
PORTLAND FL-BONNY BROOK 138 KV L	MIRA VISTA-YARNEN 345 KV	112.4	291	9.9
NO MANS LAND-VEREDA 3103 138 KV line	MICKENS-MIRA VISTA 345 KV	111.9	595	18.8
MIRA VISTA 345-1 kV XFMR	MIRA VISTA-MICKENS 345 KV	110.5	684	18.8
MICKNS-MIRA VISTA 345 KV Line	YARNEN-CALPINE KING ENERGY 345 KV	109.5	1,472	56.7
BATTLEFIELD-SYNATA HUGOTON 138 KV	MIRA VISTA-YARNEN 345 KV	109.0	665	15.0
CARROLL-CRITTENDEN 138 KV	MIRA VISTA-YARNEN 345 KV	108.6	375	10.7
ROANOKE RAPIDS 345/138 kV XFMR	MCCARTNEY-YARNEN 345 KV L	107.3	992	24.9
YARNEN-MIRA VISTA 345 KV	YARNEN-CALPINE KING ENERY 345 KV	107.3	1,859	57.9
WERNER W SUBSTATION-YAREN 138 KV line	YARNEN-MCCARTNEY 345 KV line	107.2	339	14.6
SANGER-COCHISE 138 KV	MCCARTNEY-YARNEN 345 KV line	105.2	284	6.3

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SMITHFIELD-CRITTENDEN 138 KV	MIRA VISTA-YARNEN 345 KV Line	103.3	245	8.6
YARNEN 345 kV/138 kV Transformer	MCCARTNEY-YARNEN 345 KV	102.8	381	14.6
DEL ROSA SUBSTATION-MIRA VISTA 138 KV	MIRA VISTA-MICKENS 345 KV	102.6	712	18.8
MIRA VISTA 138/1kV TX	MICKENS-MIRA VISTA 345 KV line	102.6	712	18.8
DEL ROSA-VEREDA 3103 138 KV	MICKENS-MIRA VISTA 345 KV Line	102.6	681	18.8
BOBVILLE-AVRA VALLEY SUBSTATION 138 KV	MCCARTNEY-YARNEN SUBSTATION 345 KV	101.2	445	11.2
SMITHFIELD-HANFORD 138 KV Line	MIRA VISTA SUBSTATION-YANEN 345 KV Line	101.0	256	8.7
EAST LONGMEADOW-SANGER 138 KV L	YARNEN-MCCARTNEY 345 KV L	100.5	566	11.1
PORTLAND FL-KENNETH C COLEMAN 138 KV	YAREN-MIRA VISTA SUBSTATION 345 KV Line	100.5	187	5.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TOANO 138 kV/1 kV Auto	Replace transformer with higher-rated unit	\$3,787,000	8% (\$305,000)
DANC LFGTE FACILITY-ELKO 138 KV	Rebuild line with higher-capacity conductor	\$7,682,000	8% (\$642,000)
MCCARTNEY-YARNEN 345 KV line	Rebuild line with higher-capacity conductor	\$119,520,000	8% (\$10,032,000)
FYFFE 345/138 kV TX	Replace transformer with higher-rated unit	\$12,867,000	9% (\$1,128,000)
DANC LFGTE FACILITY-BONNY BROOK 138 KV L	Rebuild line with higher-capacity conductor	\$6,358,000	8% (\$531,000)
PORTLAND FL-BONNY BROOK 138 KV L	Rebuild line with higher-capacity conductor	\$28,593,000	8% (\$2,386,000)
NO MANS LAND-VEREDA 3103 138 KV line	Rebuild line with higher-capacity conductor	\$3,962,000	8% (\$335,000)
MIRA VISTA 345 kV/1 kV TX	Replace transformer with higher-rated unit	\$13,424,000	8% (\$1,137,000)
MIRA VISTA-MICKENS 345 KV	Reconductor line	\$7,936,000	10% (\$830,000)
BATTLEFIELD-SYNATA HUGOTON 138 KV	Reconductor line	\$675,000	5% (\$36,000)
CARROLL-CRITTENDEN 138 KV	Reconductor line	\$2,094,000	8% (\$176,000)
ROANOKE RAPIDS 345 kV/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$23,166,000	8% (\$1,787,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
YARNEN-MIRA VISTA 345 KV	Reconductor line	\$27,328,000	9% (\$2,441,000)
WERNER WEST-YARNEN 138 KV	Reconductor line	\$1,958,000	9% (\$182,000)
SANGER-COCHISE 138 KV	Reconductor line	\$846,000	6% (\$49,000)
SMITHFIELD-CRITTENDEN 138 KV	Reconductor line	\$4,863,000	14% (\$657,000)
YARNEN 345/138 kV Autotransformer	Replace transformer with higher-rated unit	\$16,375,000	9% (\$1,521,000)
DEL ROSA-MIRA VISTA 138 KV	Reconductor line	\$1,735,000	8% (\$147,000)
MIRA VISTA 138-1 KV TX	Replace transformer with higher-rated unit	\$4,326,000	8% (\$366,000)
VEREDA 3103-DEL ROSA 138 KV	Reconductor line	\$1,744,000	8% (\$148,000)
BOBVILLE-AVRA VLY 138 KV	Reconductor line	\$3,445,000	7% (\$250,000)
HANFORD-SMITHFIELD 138 KV	Reconductor line	\$675,000	14% (\$92,000)
EAST LONGMEADOW-SANGER 138 KV L	Reconductor line	\$2,574,000	6% (\$160,000)
PORTLAND FL-KENNETH C COLEMAN 138 KV	Reconductor line	\$9,631,000	15% (\$1,482,000)
	Total allocated to project		\$26,820,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.